The majority of the plutonium from weapons testing was injected initially into the stratosphere. The plutonium originally in the weapon which survived the explosion would have been formed into high-fired oxide which would be expected to remain insoluble as it returned to earth. Such insoluble particles would have sunk in a rather short time into the bottom sediments of lakes, rivers, and oceans or would become incorporated in soils below the surface layer. However, in most nuclear weapon explosions a considerable amount of plutonium is generated in the explosion via ^{238}U (n,γ) reactions and subsequent α-decay of the product ^{239}U, ^{240}U, ^{241}U, etc. In total, about two thirds of the plutonium released was generated in this way. The nuclides from the (n,γ) reactions would exist as single atoms, and, hence, were never formed into high-fired oxides. The plutonium from this formation path would have been soluble and, as a result, more reactive and its behavior would be more similar to that of plutonium released from nuclear reactors, reprocessing plants and from nuclear waste repository sites.

22.5. PRESENT LEVELS OF TRU IN THE ECOSPHERE

The United Nations Scientific Committee on the Effects of Atomic Radiation (UNSCEAR), in 2000 reports data that corresponds to the following total average global nuclide depositions from atmospheric nuclear weapon tests: ^{89}Sr 11, ^{90}Sr 1.2, ^{95}Zr 19, ^{106}Ru 12, ^{137}Cs 1.8, ^{239}Pu 0.013, ^{240}Pu 0.009, and ^{241}Pu 0.278 kBq m^{-2}. Some of these radionuclides have now decayed because of their short half-lives and the end of atmospheric testing in 1980 and almost all of the ^{241}Pu has decayed to ^{241}Am.

Near test sites, reprocessing facilities, etc., the concentration of plutonium in the soil and water is much higher than in more distant locations. Generally, the great majority of plutonium is associated with sub-surface soils or sediments or with suspended particulates in water. For example, when vegetation, animals, litter and soils are compared, $\geq$99% of the plutonium is present in the soil. Similarly, in shallow bodies of water, more than 96% of the plutonium is found associated with the sediments. However, it is via the species that are soluble or attached to suspended colloids and/or particulate matter in water that plutonium is transported in the environment. Analysis of vertical plutonium migration in soils near Chernobyl and in eastern Europe from the Chernobyl accident has shown that most of the plutonium is still in the first 0.5 cm from the surface for soils with significant humic acid content. In these soils, the plutonium is mostly associated with the insoluble calcium-humate fraction. In non-humic, carbonate rich soils, the plutonium has moved several centimeters downward. Migration rates of $\leq$0.1 cm y^{-1} is associated with the humic soils and of 1–10 cm y^{-1} with the carbonate rich ones. Presumably, migration is retarded by the interaction with the immobilized humic material in soils.

Table 22.7 Concentration of plutonium in filtered samples of natural waters

Water[†]	Concentration of Pu (M)
Lake Michigan	2.0×10^{-17}
Great Slave Lake, Canada	1.5×10^{-17}
Okefenokee River, Florida	1.5×10^{-16}
Hudson River, New York	1.0×10^{-17}
Irish Sea:	
1 km from Windscale	1.6×10^{-14}
110 km from Windscale	1.1×10^{-15}
Mediterranean	2.6×10^{-18}
North Pacific (surface)	3.0×10^{-17}
South Pacific (surface)	1.0×10^{-17}

[†]Samples were passed through 0.45 μm filters.

In subsurface oxic soil near Los Alamos National Laboratory, USA, plutonium is relatively mobile and has been transported primarily by colloids in the 25−450 μm size range. Moreover, the association with these colloids is strong and removal of Pu from them is very slow. By contrast, near Sellafield in wet anoxic soil, most of the Pu is quickly immobilized in the sediments although a small fraction remain mobile. Differences in oxidation state (Pu(V) vs. Pu(IV)) as well as in humic content of the soils may explain these differences in mobility.

Table 22.7 lists the concentration of plutonium, after filtration (0.45 μm), in the surface layers of some natural waters. The higher concentration in the Okefenokee River is assumed to reflect the effect of complexing by humic materials. This agrees with the observation that adding humic material to seawater samples containing plutonium increases the solubility by more than a factor of five over a period of one month.

It is difficult to obtain reliable values of plutonium concentration in natural aquatic systems as it is very low, approximately 0.001 dpm per liter sea water. Moreover, the plutonium associated with suspended particles may be more than an order of magnitude greater than that in true solution. In tests of water from the Mediterranean Sea, filtration (0.45 μm) reduced the concentration of plutonium by a factor of 25. In laboratory tests with filtered seawater to which plutonium was added, after one month the total concentration of Pu was 1.3×10^{-11} M, but only 40% (5×10^{-12} M) was in solution as ionic species and the other 60% was probably in colloidal form. The mean residence time of Pu in the water column is proportional to the concentration of particulate matter. As a consequence, >90% of the Pu is rapidly removed from coastal waters whereas, in mid-ocean waters where the particulate concentrations are lower, the residence time for Pu is much longer. It also means that Pu is accumulated in clays and silt near reprocessing plant effluent release points. Known cases are near Sellafield, and in the river downstream from the old norvegian reprocessing plant at Lilleström.

22.6. ACTINIDE CHEMISTRY IN THE ECOSPHERE

22.6.1. Redox properties

Before proceeding to more detailed discussion of the behavior of actinides in the environment, it is useful to review some of their chemical properties. A general discussion of the actinide solution chemistry is given in §14.3; here, the focus is on their behavior in aqueous solutions and primarily in solutions of pH 5–9 which is the pH range of natural waters (e.g. the oceans have pH = 8.2). The actinide ions have an unusually broad range of oxidation states in aqueous solution, from II to VII; The II and VII States are not discussed further as they do not form in ecosystems. Following the normal pattern for polyvalent cations, lower oxidation states are stabilized by more acidic conditions while higher oxidation states are more stable in basic solutions. Of course, this generalization can be negated by other factors, such as complexing, which may cause a reversal of the relative stability of different oxidation states. The greater strength of complexing of An(IV) cations relative to that of An(III) can significantly increase the apparent redox stability of the An(IV) species compared to An(III). The greater tendency to hydrolysis of Pu(IV) causes Pu(III), which is stable in acid solution, to be oxidized to Pu(IV) in neutral media. The disproportionation of Pu(V) is discussed in §14.3.3, where it is pointed out that in the higher pH and very low concentrations of Pu in natural waters, disproportionation is not a factor in the redox behavior of plutonium.

The actinide elements in a particular oxidation state (e.g. Th(IV), U(IV), Pu(IV), Np(IV), Am(IV)) have similar behavior. However, their redox behavior is quite different, as mentioned in Chapter 14. The pH affects this redox behavior significantly as reflected in Figure 22.5 which compares the redox potentials of U, Np and Pu at pH = 0, 8 and 14.

Am (III) is the most stable oxidation state in aqueous solutions while Pu(III) and Np(III) are present under reducing conditions (e.g. anoxic waters). Th(IV) is the common and stable state for that element. U(IV) and Np(IV) do not react with water but are oxidized by O_2 in oxic systems. Pu(IV) is stable at low concentrations in acidic solution, but $Pu(OH)_4$ has a very low solubility product.

NpO_2^+ is stable except at high acidities and high concentrations under which conditions it disproportionates. UO_2^+ and PuO_2^+ increase in stability as the pH is increased. U, Np and Pu form AnO_2^{2+} ions in solution with the stability decreasing in the order U > Pu > Np. UO_2^{2+} is the most stable uranium species in natural waters.

Redox properties are often described by the aid of potential-pH (or Pourbaix) diagrams, Figure 22.6. The shaded area represents typical groundwaters in granitic rock, containing iron minerals; such waters are usually reducing, i.e. have *E*h-values below 0. Natural groundwaters (including oceans, lakes, rivers, etc) fall within the area enclosed by the dashed curve; they can have rather high *E*h values due to atmospheric oxygen, and also be rather alkaline in contact with carbonate rocks. The sloping lines follow the Nernst equation (defined by eqn. (18.6)). We discuss this Figure further in §22.7.1.